

Five-Minute Piano Drills

Twelve drills to repeat every day, from counting beats to writing your own melodies

What this set is for

Three targets:

- Play **The Scientist** (Coldplay, F major, slow) and **The Loneliest Girl** (Carole & Tuesday, G major, 112 BPM) with confidence.
- Understand scales, chords, bass and melody well enough to **write your own** lo-fi piano loop or a simple synth melody.
- Build the hand skills underneath: steady rhythm, both hands at once, moving between chords without looking.

Drum programming is left out. Everything here happens at the keyboard.

These are drills, not lessons

You don't finish a drill. You repeat it for weeks and it gets easier and faster, the same way a gym routine does. Every drill has a **level ladder**. Climb it slowly.

The daily routine

1. **Set a 5-minute timer.** Stop when it rings, even if it's going well. Tomorrow you'll want more.
2. **One front drill** (the lowest drill you haven't mastered). If you have 10 minutes, add **one review drill** from earlier.
3. Follow the "**The 5 minutes**" plan on the page. The last minute is always free playing. Don't skip it; it's the part that makes you come back.

Three rules

1. **Count out loud.** "One and two and..." Every time. It fixes more mistakes than anything else.
2. **Metronome ladder.** Start at a tempo where you make no mistakes. After 3 clean runs, add 5 BPM. Two stumbles in a row: drop 5 BPM.
3. **Slow and clean beats fast and messy.** Your hands learn whatever you repeat, including the mistakes.

When to move on, when to go back

Move on when you tick the level the page marks as the gate (usually L3 or L4). You don't need the top levels before starting the next drill. Keep climbing them as review.

Go back when a later drill falls apart. It is nearly always a gap in an earlier one: shaky chords (3–4), weak left hand (5) or lost count (1). Spend a few days there. That's normal.

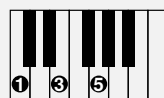
Kit

Metronome app. Phone voice recorder, or a DAW with MIDI recording. Hearing yourself back, or seeing your notes in a piano roll, is the fastest feedback you can get. Sustain pedal if you have one.

Reference sheets

Sheets **2+** and **3+** come straight after Drills 2 and 3. They aren't drills: they list the exact notes of every key and every chord. Keep them on the music stand.

How to read the pages



● black dot = press this key. The number is a **finger** or a **scale step**; the page tells you which.

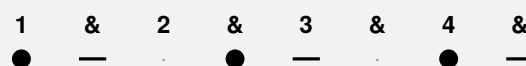
▲ triangle = the **root** or home note.

Fingers: 1 = thumb, 2 = index, 3 = middle, 4 = ring, 5 = pinky. Same on both hands.

Chord symbols: C = C major, Am = A minor, Dm7 = D minor seventh, Fsus2 = F with G instead of A, Bb° = diminished.

Roman numerals number the chords of a key: I = the chord on step 1, vi = the chord on step 6. Capitals are major, small letters are minor. Drill 3 explains it.

Rhythm grids split a 4/4 bar into 8 slots (every drill here is in 4/4):



● = play, — = keep holding, · = nothing new. This one is the 3-3-2 rhythm of Drill 1.

The map

#	Drill	What it gives you
1	Count the Groove	steady time, syncopation, the pop "push"
2	The Major Scale	keys of G and F, fingering, steps and thirds
3	Chord Staircase	every chord in a key, the number system
4	Lazy Hands	inversions; both songs' chord loops
5	The Bass Engine	left-hand bass patterns, 8th-note drive
6	Broken Chords	arpeggios, lo-fi texture
7	Split the Beat	hands together, the pop groove
8	Lo-fi Colours	7ths, sus2, add9
9	No Wrong Notes	pentatonic improvising, target notes
10	Build a Tune	motif to 8-bar melody
11	Big Five Progressions	the loops under most songs
12	Chord Vocabulary	the song chords, nine chord types

Typical path

Weeks 1–3: Drills 1 + 2. Weeks 3–6: 3 + 4, with the song chord loops appearing. Weeks 6–10: 5, 6, 7 (both hands, grooves). From there: 8, 9, 10 (colour and your own melodies), while the song roadmap on the last pages pulls it together. Slower is fine.

Drills 11 and 12 run alongside. Start 12 once Drill 3 passes its gate and 11 once Drill 4 does, and use them as your review drill on the days you have 10 minutes.

DRILL 1

Count the Groove

One chord, six rhythms. Rhythm first, because everything sits on it.

Needs: nothing **Skills:** pulse, 8th notes, syncopation, the "push" **Gate to Drill 2+:** L3

Why this matters

Music runs on a grid. In 4/4 each bar has **4 beats**, and each beat splits in two: **1 & 2 & 3 & 4 &**. That makes 8 slots per bar, and every rhythm in pop, lo-fi and EDM is a choice of which slots to play. Notes on the numbers feel steady. Notes on the "&"s feel like they lean forward; that is **syncopation**. The 3-3-2 pattern (C below) sits under a huge amount of pop and electronic music. Pattern F plays the next chord half a beat early. That early chord is called a **push**, and pop piano like The Loneliest Girl is full of them.

The six patterns

	1	&	2	&	3	&	4	&
A Quarters	●	.	●	.	●	.	●	.
B Eighths	●	●	●	●	●	●	●	●
C 3-3-2	●	.	.	●	.	.	●	.
D Charleston	●	.	.	●	.	.	.	.
E Off-beats	.	●	.	●	.	●	.	●
F The push	●	.	.	.	●	.	.	●

How to play a pattern: right hand plays the chord on every ●, left hand holds the root (bottom note) for the whole bar.

E: this is the house-piano off-beat stab. Say the numbers, play only on the "&"s.

F: the hit on 4& is *already the next bar's chord*. Hold it through the next beat 1 and don't replay it. From bar 2 on, beat 1 stays silent.

The four chords

Right hand fingers 1-3-5, left hand plays the single low note. Here with pattern C (3-3-2); counts written under.

The 5 minutes

0:00–1:00

Clap and count. Metronome at 60–70. Clap pattern A, then B, then C, 2 bars each, counting out loud.

1:00–3:00

Today's two patterns on C. Pick two patterns from the grid. Play each for 4 bars on C major. Count out loud the whole time.

3:00–4:00

Add the chord loop. Same patterns over C – Am – F – G, one chord per bar, round and round.

4:00–5:00

Play. Keep the loop going and switch patterns whenever you like. That is a pop song's backing.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	Patterns A, B, C on one chord, counting aloud, no stops	60 BPM	_____
☐ L2	All six patterns on one chord	70	_____
☐ L3	Gate. Any pattern over C–Am–F–G, 8 bars without a break	70	_____
☐ L4	Switch pattern every 4 bars without stopping (A→C→E→F...)	80	_____
☐ L5	Left hand plays the root on beats 1 and 3 while the right plays the pattern	80	_____
☐ L6	Pattern C and F at The Loneliest Girl's tempo	112	_____
☐ L7	Roll a die each bar (1–6 = A–F) and play what it says	90+	_____

Make it harder

Close your eyes. Metronome clicks only on beats 2 and 4. Record into a DAW, open the piano roll and see how close to the grid you really were. Invent pattern G.

Song link

Loneliest Girl: pushes (F) at 112 BPM. **The Scientist:** steady 8ths (B) in the left hand, slow. **EDM:** off-beats (E) and 3-3-2 (C) are the basic moves.

Seven notes, one step pattern. Learn it once and it works in every key.

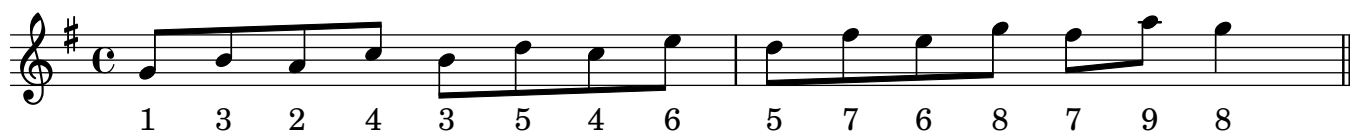
Needs: nothing (pairs with Drill 1) **Skills:** keys of G and F, fingering, scale steps, thirds **Gate:** L4

A **key** is a set of 7 notes. A melody in G major takes almost all of its notes from the G major scale. Every major scale uses the same step pattern: **whole, whole, half, whole, whole, whole, half** (W-W-H-W-W-W-H). A half step is the next key, black or white. A whole step skips one. Start the pattern on C and you get only white keys. Start on G and you need F# to keep it right. Start on F and you need Bb. That is why **G major has one sharp** (The Loneliest Girl) and **F major has one flat** (The Scientist). Number the notes **1 to 7**. Step 1 is “home”, where a melody sounds finished. Chords, bass lines and melodies in the later drills are all described with these numbers. **The exact notes of each key are on Sheet 2+, the next page.**

The image shows two musical staves. The top staff is for G major (one sharp, treble clef, common time) and the bottom staff is for F major (no sharps or flats, treble clef, common time). Both scales are written in a single line with fingerings indicated by numbers 1-5. The G major scale is: G4 (1), A4 (2), B4 (3), C5 (1), D5 (2), E5 (3), F#5 (4), G5 (5), F#5 (4), E5 (3), D5 (2), C5 (1), B4 (3), A4 (2), G4 (1). The F major scale is: F4 (1), G4 (2), A4 (3), Bb4 (4), C5 (1), D5 (2), E5 (3), F5 (4), E5 (3), D5 (2), C5 (1), Bb4 (4), A4 (3), G4 (2), F4 (1).

Walking in thirds (skip one note each time)

Walking in thirds (skip one note each time)



The 5 minutes

0:00-1:00

C major, right hand, up and down. Say the step numbers out loud: “1 2 3 4 5 6 7 1”.

1:00–2:30

Key of the day. G on odd days, F on even days. Right hand, then left hand, slowly and evenly.

2:30–4:00

Thirds walk in the key of the day (above). This is how chords are built in Drill 3.

4:00–5:00

Play. Left hand holds the note on step 1 down low. Right hand plays any notes from today's scale in any order. None of them will sound wrong, because they're all in the key. Finish on 1 and listen to it land.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

- | | | | |
|--|---|----|-------|
| ☐ L1 | C, G and F, right hand, one octave, even quarter notes | 60 | _____ |
| ☐ L2 | Same with the left hand | 60 | _____ |
| ☐ L3 | Both hands as 8th notes | 60 | _____ |
| ☐ L4 Gate. | Hands together, one octave, G and F; thirds walk in both keys | 60 | _____ |
| ☐ L5 | Two octaves, hands together, 8ths | 80 | _____ |
| ☐ L6 | Add D major (F# C#) and Bb major (Bb Eb). Work out the notes yourself with W-W-H-W-W-W-H | 70 | _____ |
| ☐ L7 | Minor keys: G major's notes from E to E is E minor ; F major's from D to D is D minor | 70 | _____ |

Play in rhythm pattern C or F from Drill 1. Play step patterns 1-2-3-1, 2-3-4-2...up the scale. Hands in opposite directions (start on the same note, move apart).

The melodies of both songs come almost entirely from these two scales. The **E minor** and **D minor** in L7 matter too: both songs keep leaning on their minor chord (Em and Dm).

Scale Sheet

The exact notes of every key you'll need. Keep it next to Drill 2.

Use with: Drills 2, 9 and 10 Answers: "Which notes am I allowed to play in this key?"

Three rules that answer "which notes?"

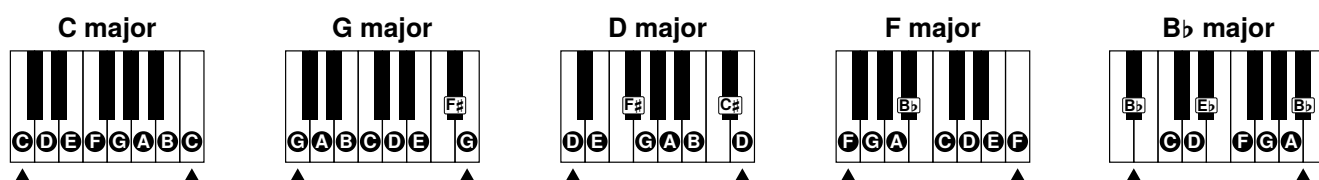
1. A key uses exactly 7 notes, one of each letter **A B C D E F G**, in order. A scale never skips a letter and never uses a letter twice. G major can't contain both F and F#, so its 7th note is F# *instead of* F.
2. The step pattern decides which letters get a # or b: W-W-H-W-W-W-H (whole, whole, half, whole, whole, whole, half). The keyboard already has two built-in half steps with no black key between them: **E-F** and **B-C**. When the pattern needs a whole step there, or a half step somewhere else, a black key fills in.
3. Once you know the 7 notes, those are the only notes of the key. Improvise, write melodies and build chords (Sheet 3+) from that row only. Anything else is "outside" and sounds tense unless you mean it.

Worked example: building D major

D $\xrightarrow{W}$ E $\xrightarrow{W}$ F# $\xrightarrow{H}$ G $\xrightarrow{W}$ A $\xrightarrow{W}$ B $\xrightarrow{W}$ C# $\xrightarrow{H}$ D
 1 2 3 4 5 6 7 1

E to F is only a half step, but the pattern needs a whole step, so step 3 becomes F#. B to C is a half step, but steps 6 to 7 need a whole step, so step 7 becomes C#. That's why D major has two sharps. Every key works the same way.

The keys on the keyboard (black dots = the 7 notes; triangle = step 1)



The eight keys worth knowing (the notes with # or b are in bold)

Key	Signature	1	2	3	4	5	6	7	Same notes as	Used in
C major	none	C	D	E	F	G	A	B	A minor	Drills 1, 4, 10
G major	1#: F	G	A	B	C	D	E	F#	E minor	The Loneliest Girl
D major	2#: F C	D	E	F#	G	A	B	C#	B minor	
A major	3#: F C G	A	B	C#	D	E	F#	G#	F# minor	
E major	4#: F C G D	E	F#	G#	A	B	C#	D#	C# minor	
F major	1b: B	F	G	A	Bb	C	D	E	D minor	The Scientist
Bb major	2b: B E	Bb	C	D	Eb	F	G	A	G minor	
Eb major	3b: B E A	Eb	F	G	Ab	Bb	C	D	C minor	

Sharps always arrive in the order **F C G D**, flats in the order **B E A D**. "Same notes as" is the key's relative minor: the same 7 notes, with home on step 6 instead of step 1.

Reading the key signature

The # or b signs at the start of every line are the key signature. A # on the F line means *every* F in the piece is F#, in every octave, unless a note is marked otherwise.



How to use this in Drill 2

Pick today's key from the table. Before playing, say its 7 notes out loud ("G A B C D E F#"). Then play only those. If a note sounds wrong in the free-play minute, check it against the row.

Self-test

Cover the table. Build A major and Eb major with W-W-H-W-W-W-H, the way D major is built above, then check. When you can do any row in under a minute, you know how keys work.

Chord Staircase

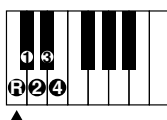
Stack thirds on every scale note and you get the seven chords of the key.

Needs: Drill 2 L4 **Skills:** intervals, major/minor triads, the number system **Gate:** L4

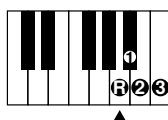
Why this matters

An **interval** is a distance counted in half steps. The one that matters most is the **third**: **major third = 4 half steps** (bright), **minor third = 3 half steps** (darker). A **triad** is two thirds stacked: root + 3rd + 5th, which is “skip a note, skip a note” in the scale. Build a triad on every step of a major scale, using only notes of that scale, and you always get the same pattern: **major, minor, minor, major, major, minor, diminished**. Name them with Roman numerals: I ii iii IV V vi vii°. When producers say “it’s a 6-4-1-5”, this is what they mean. The numbers work in any key, so one progression can be moved anywhere. **The notes of every chord, sharps and flats included, are on Sheet 3+, the next page.**

Major 3rd: C to E = 4 half steps



Minor 3rd: A to C = 3 half steps



Major chord = major 3rd, then minor 3rd on top.

Minor chord = minor 3rd, then major 3rd.

The staircase in G (right hand, fingers 1-3-5 for every chord)

Diagram showing the chord staircase in G major (right hand, fingers 1-3-5 for every chord). The chords are G, Am, Bm, C, D, Em, F#°, and G. The corresponding Roman numerals are I, ii, iii, IV, V, vi, vii°, and I.

	I	ii	iii	IV	V	vi	vii°
C major	C	Dm	Em	F	G	Am	B°
G major	G	Am	Bm	C	D	Em	F#°
F major	F	Gm	Am	Bb	C	Dm	E°

The 5 minutes

- 0:00–1:00** **Hear the thirds.** On white keys play C–E, D–F, E–G, F–A... together. Say “bright” (major 3rd) or “dark” (minor 3rd) for each, then count half steps to check.
- 1:00–2:30** **Staircase** up and down in C, then G or F. Say each chord’s name and number. Left hand adds the root, low.
- 2:30–4:00** **Number game.** Say a random number from 1 to 6, find that chord in G within 2 seconds, then do the same in F.
- 4:00–5:00** **Play the songs from numbers.** In G: **vi – IV – I – iii** = Em – C – G – Bm: the verse of *The Loneliest Girl*. In F: **vi – IV – I – I** = Dm – Bb – F – F: *The Scientist*. Loop each 4 times, slowly.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	Staircase in C, up and down, naming chords aloud	60
☐ L2	Staircase in G and F	60
☐ L3	Left-hand root with every chord	60
☐ L4	Gate. Number game: any chord in G or F in under 2 seconds	—
☐ L5	Staircase in D and Bb; write the table row yourself first	60
☐ L6	Broken staircase: each chord as root-3rd-5th-3rd in 8ths	70
☐ L7	Say any 4 numbers (e.g. 1-5-6-4) and play them in any key you know	—

Make it harder

Close your eyes for the number game. Left hand plays the staircase too. Find the vi chord of every key you know: that is the key’s **relative minor**.

Song link

Both songs are 4-chord loops written in this number system. The iii chord (Bm in G) is what gives *The Loneliest Girl*’s verse its bittersweet sound. Most pop uses I, IV, V and vi only.

Chord Sheet

The exact notes in every chord, and why some of them are sharp or flat.

Use with: Drills 3, 4, 11 and 12 Answers: "Is it F or F# in this chord?"

Finding a chord's notes in three steps

1. Letters first: skip a letter, twice. Every triad is three letters, each one skipping the next. There are only seven letter patterns: **C-E-G D-F-A E-G-B F-A-C G-B-D A-C-E B-D-F.**

2. Apply the key's sharps or flats (Sheet 2+). *Inside a key, a chord uses only that key's notes.* In G major every F is F#, so D-F-A becomes **D F# A**, B-D-F becomes **B D F#**, and F-A-C becomes **F# A C**. In F major every B is Bb, so B-D-F becomes **Bb D F** and G-B-D becomes **G Bb D**.

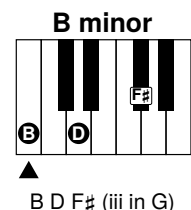
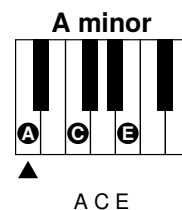
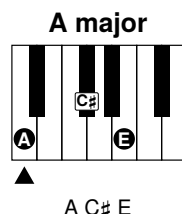
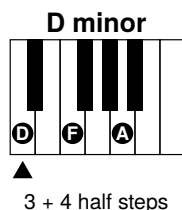
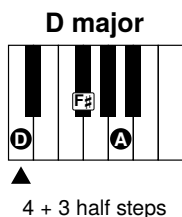
3. A chord name fixes its notes everywhere. D major is *always* D F# A, in any song and any key, because major means 4 + 3 half steps. Minor is 3 + 4. If a chart shows a chord that isn't in the key, trust the chord name and use the table at the bottom.

The chords of each key, with their notes (sharps and flats in bold)

	I	ii	iii	IV	V	vi	vii°
C major	C C E G	Dm D F A	Em E G B	F F A C	G G B D	Am A C E	B° B D F
G major	G G B D	Am A C E	Bm B D F#	C C E G	D D F# A	Em E G B	F#° F# A C
D major	D D F# A	Em E G B	F#m F# A C#	G G B D	A A C# E	Bm B D F#	C#° C# E G
F major	F F A C	Gm G Bb D	Am A C E	Bb Bb D F	C C E G	Dm D F A	E° E G Bb
Bb major	Bb Bb D F	Cm C Eb G	Dm D F A	Eb Eb G Bb	F F A C	Gm G Bb D	A° A C Eb

The same chord can live in several keys: Em is vi in G, iii in C and ii in D, and its notes are always E G B. Chords made only of white keys (C, Dm, Em, F, G, Am) turn up in many keys.

Major or minor: only the middle note moves, by one half step



Every major and minor chord (use it when a chord isn't in your key)

Root	Major	Minor
C	C E G	C Eb G
C# / Db	Db: Db F Ab	C#m: C# E G#
D	D F# A	D F A
Eb	Eb G Bb	Eb G b Bb
E	E G# B	E G B
F	F A C	F Ab C

Root	Major	Minor
F#	F# A# C#	F# A C#
G	G B D	G Bb D
Ab / G#	Ab: Ab C Eb	G#m: G# B D#
A	A C# E	A C E
Bb	Bb D F	Bb Db F
B	B D# F#	B D F#

How to use this in Drill 3

Before each staircase chord, say its three letters out loud ("B, D, F-sharp"), then play it. Saying the notes is what makes them stick. After a week, try it without looking at the table.

Quick check

Why is there no F# in any chord of C major? (C major has no sharps.) D and Dm both turn up in songs: which note tells them apart? (The middle one: F# in D, F in Dm.)

Lazy Hands

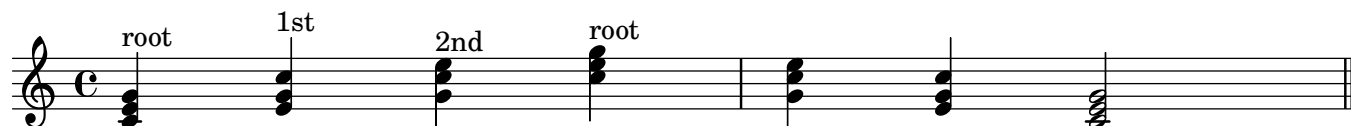
Inversions: move each finger to the nearest note instead of jumping the whole hand.

Needs: Drill 3 L4 **Skills:** inversions, voice leading, both songs' chord loops **Gate:** L4

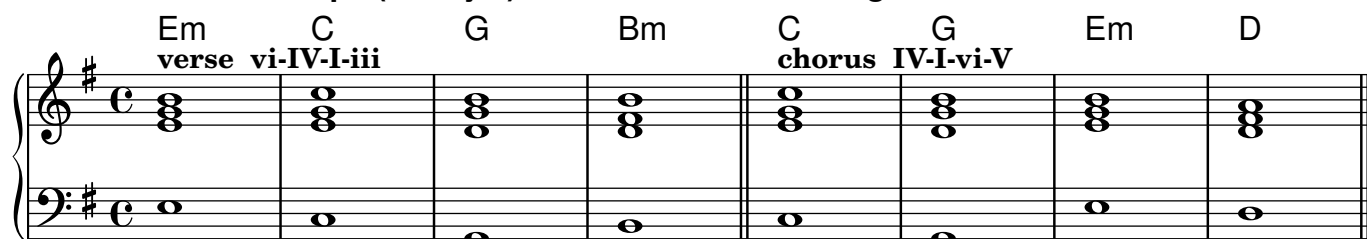
Why this matters

You can stack a chord's three notes in any order. **Root position** has the root at the bottom (C-E-G). **1st inversion** puts the 3rd at the bottom (E-G-C). **2nd inversion** puts the 5th at the bottom (G-C-E). All three are still C major. Jumping from one root-position chord to the next sounds blocky and makes your hand leap. Pianists instead **keep notes that two chords share, and move the other notes by the smallest step**. This is **voice leading**, and it's why good piano parts sound smooth. In both songs the right hand barely moves, because the chords are voiced this way. The left hand plays the root, so the chord name stays clear.

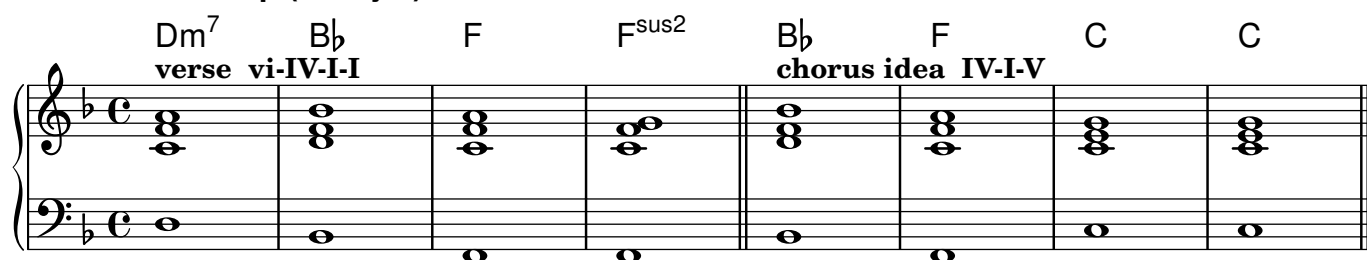
Inversion climb



The Loneliest Girl loops (G major). Watch how little the right hand moves.



The Scientist loop (F major)



Right hand C-F-A over a D in the bass is **Dm7**. The F^{sus2} bar swaps A for G: the top finger steps down and back up. Listen for it.

The 5 minutes

0:00–1:00

Inversion climb on C, then G, then F. Say “root, first, second” as you go.

1:00–2:30

Loneliest Girl verse loop, one chord per bar, left hand on the root. Look at the *sheet*, not your hands.

2:30–4:00

Find it yourself. Take G – D – Em – C (I–V–vi–IV). Start with G as G-B-D and find the nearest shape of each next chord. Aim for a top note that moves by a step or not at all.

4:00–5:00

Play The Scientist loop slowly (about 60), with the sustain pedal changed on each chord. It will sound like the record.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	Inversion climb in C, G, F	60	_____
☐ L2	Climb every chord of the G staircase (Drill 3) in all three positions	60	_____
☐ L3	Loneliest Girl verse + chorus loops, 4× each, eyes on the page	70	_____
☐ L4	Gate. Scientist verse + chorus idea, 4× each, with pedal	60	_____
☐ L5	Start each loop from a different shape of the first chord (3 versions of every loop)	70	_____
☐ L6	Play the loops without the page, then with your eyes closed	80	_____
☐ L7	Move the Loneliest Girl verse (vi-IV-I-iii) to C and to F, voicing it yourself	70	_____

Make it harder

Two chords per bar. Left hand plays root *octaves*. Play the loops in Drill 1 patterns C and F.

Song link

This page *is* the harmony of both songs. Drills 5–8 only add rhythm and colour on top.

The Bass Engine

Six left-hand patterns: the bass line of pop songs and electronic tracks.

Needs: Drill 4 L3 **Skills:** left-hand independence, bass patterns, 8th-note drive **Gate:** L4

Why this matters

The **bass** is the lowest line, and it does two jobs. It tells the ear **which chord** is playing (usually by playing the root), and it **drives the rhythm**. In a track, bass and kick drum lock together into the groove. On piano the left hand is the bass player. Once it can run a pattern on autopilot, the right hand is free for chords and melody. The same patterns carry straight over to a bass synth: whole-note roots are a sub-bass, 8th-note roots are the Coldplay/rock drive, octaves are disco and French house, off-beats are trance and house.

Six patterns (shown on D; follow the chord roots when the chord changes)

Fingers: B3 is 5 on the low note, 1 on the high note. B4 is 5-2-1-2. **B6:** on the last 8th, step to the note next to the *next* root, so the bass line leads into the chord change.

The practice loop: Dm – B♭ – F – C (F major, vi–IV–I–V)

Left hand roots: D – B♭ – F – C, one per bar. Right hand either rests or holds the Drill 4 chord shapes.

The 5 minutes

0:00–1:00

B1 and B2 around the loop, metronome at 70. Make every 8th exactly the same volume.

1:00–3:00

Pattern of the day (B3, B4, B5 or B6) around the loop, 4 rounds. Count out loud.

3:00–4:00

Hands together, minimum version. The left hand runs the pattern. The right hand plays one whole-note chord per bar, using the shapes from Drill 4.

4:00–5:00

Play. Mix patterns: 4 bars of B2, then 4 of B4, then B5 ... You're a one-person band.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	B1, B2 around the loop, left hand alone	70	_____
☐ L2	B3 and B4, left hand alone	70	_____
☐ L3	B5: metronome clicks are the kick, you play between them	90	_____
☐ L4	Gate. B2 + right-hand whole-note chords, Scientist loop	73	_____
☐ L5	Loneliest Girl verse (Em–C–G–Bm) with B2 or B3	100–112	_____
☐ L6	B6 approach notes into every chord change	80	_____
☐ L7	Invent your own 1-bar bass rhythm and loop it 8 bars under chords	any	_____

Make it harder

Swap B4's 8ths for 3-3-2 (Drill 1, pattern C). Play B2 staccato, then legato. Record the left hand into a DAW on a bass synth and loop it: that's a track foundation.

Song link

The Scientist: soft, even B2 8ths is the engine of the song.
Loneliest Girl: roots and octaves on the beat at 112. For more on bass and kick see *Making Music* (DeSantis), "Bass Lines and Kick Drums as a Single Composite", p. 158.

Broken Chords

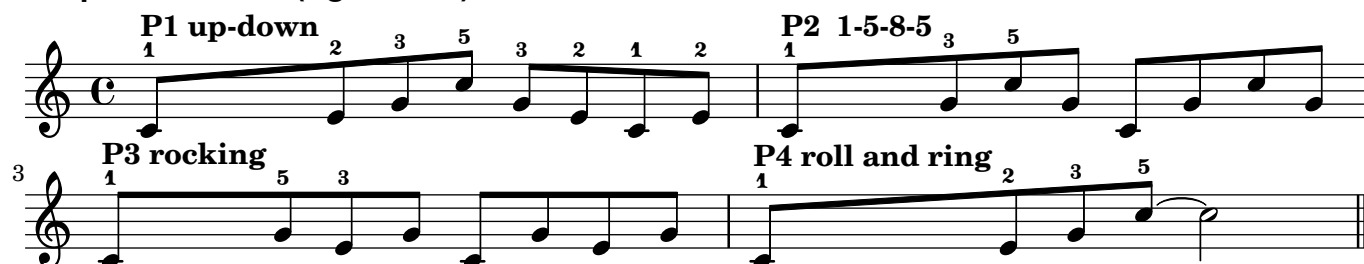
Play the chord one note at a time and it starts to move. Lo-fi and ballad texture.

Needs: Drill 4 L4 **Skills:** right-hand arpeggios, even 8ths, swing feel **Gate:** L4

Why this matters

A chord doesn't have to be struck all at once. Play its notes one after another (an **arpeggio**, or broken chord) and a single held harmony gets motion and a pulse. Piano ballads, lo-fi loops and the synth **arpeggiator** in your DAW all do this. The arpeggiator is exactly this drill, automated. Keep the hand in one shape and let the fingers do the work. Use the Drill 4 voicings so the shape barely changes between chords. **Swing:** lo-fi and hip-hop play 8ths long-short ("daa-da daa-da") instead of even. It is the quickest way to make a loop sound lo-fi.

Four patterns on C (right hand)



Loop: Dm7 – B♭ – F – C with P3, using Drill 4 shapes



The 5 minutes

- 0:00–1:00** P1 on C, then on Am, F, G (C – Am – F – G). Even 8ths, metronome 60–70.
- 1:00–3:00** Pattern of the day (P2, P3 or P4) over the loop above, with the left hand on whole-note roots.
- 3:00–4:00** Same pattern, swung. Long-short, long-short. Slow down to 60–70.
- 4:00–5:00** Lo-fi minute. Tempo about 70, sustain pedal changed on each chord, soft touch, swing on. Mix P3 and P4. Record it: this is a usable lo-fi loop.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	P1 over C–Am–F–G, even	70	_____
☐ L2	All four patterns, one chord each	70	_____
☐ L3	P3 over the Dm7 loop above, hands together	70	_____
☐ L4	Gate. Any pattern over both songs' loops (Drill 4 shapes)	70	_____
☐ L5	Swing, and change pattern every 2 bars	80	_____
☐ L6	16th notes: each pattern twice as fast	60	_____
☐ L7	Left hand B2 8ths (Drill 5) + right-hand arpeggio	70	_____

Make it harder

Invent a P5 from 4 chord notes in a new order (for example 3-1-5-1). Let the top note of each bar form a little tune. Play the arpeggio in the left hand instead.

Song link

Broken chords fill the space when the melody rests in **The Scientist**. In a DAW, compare your P1 with the arpeggiator's "up-down" mode.

DRILL

7

Split the Beat

Two hands, two rhythms. The pop-piano groove.

Needs: Drills 1 (L5), 4 and 5 (L4)

Skills: hand independence, two-hand syncopation

Gate: L5

Why this matters

Hand independence isn't two brains. The trick is to learn **both hands as one combined rhythm**: on each slot of the bar, know whether it's left, right, both or nothing. The basic pop-piano recipe: **left hand keeps the beat** (roots on 1 2 3 4), **right hand syncopates** (3-3-2 hits, pushes). That tension between steady and off-beat is what makes it groove.

The four stages (L = left only, R = right only, B = both together)

	1	&	2	&	3	&	4	&	
S1 Together	B	.	L	.	B	.	L	.	right hand on 1 and 3
S2 Double	B	R	B	R	B	R	B	R	right-hand 8ths
S3 3-3-2	B	.	L	R	L	.	B	.	Drill 1 pattern C
S4 The push	B	.	L	.	B	.	L	R	4& = next chord, held

S3 on The Loneliest Girl verse (G major)

Em

C

G

Bm

S4: the push (right hand arrives half a beat early and holds across the bar line)

Em

C

G

Bm

The 5 minutes

- 0:00–1:00

Tap it first. Tap today's stage on your knees (left knee = left hand). Say the counts.
- 1:00–3:00

Play today's stage on Em – C – G – Bm. Start at 60. Stop and slow down if the left hand drifts off the beat.
- 3:00–4:00

Same stage on the chorus loop C – G – Em – D (shapes from Drill 4).
- 4:00–5:00

Play. Verse 4 bars, chorus 4 bars, any stages you like. Hum along.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row		
☐ L1	S1 and S2 on the verse loop	70
☐ L2	S3 on the verse loop	60
☐ L3	S3 verse + chorus, no stops	80
☐ L4	S4 verse + chorus	70
☐ L5	Gate. S3 or S4, verse + chorus	100
☐ L6	Full Loneliest Girl tempo	112
☐ L7	Scientist texture: left hand B2 8ths, right hand held chords that change on the push	73

Make it harder

Left hand plays B3 octaves instead of quarters. Right hand plays S3 while the left does B5 off-beats. Sing the counts instead of saying them.

Song link

This is the accompaniment for **The Loneliest Girl**. Once S4 is smooth at 112, the song's backing is in your hands. Adding its melody is the step after (see the song roadmap page).



Lo-fi Colours

Add one more note: 7ths, sus2 and add9 turn plain chords into lo-fi chords.

Needs: Drills 3 and 4 (L4) **Skills:** 7th chords, sus and add9, rich voicings **Gate:** L4

Why this matters

Stack one more third on a triad and you get a **7th chord**. Three kinds matter: **maj7** (dreamy, on I and IV), **m7** (soft, mellow, on ii, iii and vi) and **dominant 7** (tense, wants to move, on V). **sus2 / sus4** replace the 3rd with the 2nd or 4th. The chord then sounds open, neither happy nor sad. **add9** keeps the triad and adds the 9th (step 2, an octave up). Lo-fi uses mostly maj7, m7 and 9ths, played slowly and softly with swing.

The shortcut: left hand plays the root, right hand plays **the triad that starts on the 3rd**. Dm7 = D + F major triad. Fmaj7 = F + A minor triad. You already know every triad from Drill 3.

	I maj7	ii7	iii7	IV maj7	V7	vi7	viiø
C major	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bm7b5
F major	Fmaj7	Gm7	Am7	Bbmaj7	C7	Dm7	Em7b5
G major	Gmaj7	Am7	Bm7	Cmaj7	D7	Em7	F#m7b5

Lo-fi loop in C: IVmaj7 – iii7 – ii7 – Imaj7 (right hand plays a triad, left hand the root)

Scientist colours (F major): the notes F and C ring through every chord

The 5 minutes

0:00–1:00

7th staircase in C. Play each triad, then add the 7th on top. Say “dreamy”, “soft” or “tense” for each.

1:00–3:00

Lo-fi loop above, 4 rounds. Then arpeggiate it with a Drill 6 pattern, swung.

3:00–4:00

Scientist colours. Compare Dm7 with a plain Dm, and Bbadd9 with a plain Bb.

4:00–5:00

Lo-fi minute. Tempo 70, pedal, soft. Pick a loop and hold each chord 2 bars. Improvise slowly with the top note.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	Build all seven 7ths in C by stacking thirds	—
☐ L2	Lo-fi loop, using the triad-on-the-3rd shortcut	70
☐ L3	Scientist colours loop	60
☐ L4	Gate. Loneliest Girl verse as Em7 – Cmaj7 – Gmaj7 – Bm7 in G	70
☐ L5	ii7–V7–Imaj7 in C, F and G (Dm7–G7–Cmaj7 ...)	70
☐ L6	9th chords: Cmaj9 = C + Em7 on top. Find Fmaj9 and Dm9	—
☐ L7	Any loop, arpeggiated and swung, plus Drill 5 bass	75

Make it harder

Play sus4 then resolve to the plain chord (Csus4 → C), as The Scientist’s chorus does on C. Invert the right-hand triads (Drill 4) so the top note moves by step.

Song link

Chord charts of **The Scientist** are full of sus2 and add9: that’s this page. For **lo-fi** this page is most of the harmony you’ll need.

No Wrong Notes

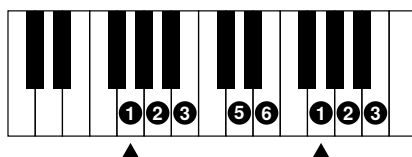
Improvise with 5 safe notes, then learn to aim for the right one.

Needs: Drills 2 and 5 (L4) **Skills:** pentatonic scale, target notes, phrasing **Gate:** L4

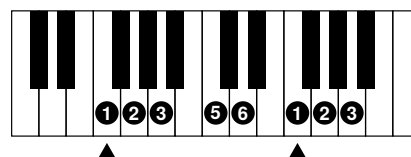
Why this matters

The **major pentatonic** is the major scale without steps 4 and 7, which leaves **1 2 3 5 6**. Steps 4 and 7 sit a half step from other notes and can clash against the chords. Take them out and almost anything you play over the I, IV, V and vi chords sounds fine. It's the safest scale there is for making up melodies. Start the same five notes on step 6 and you have the **minor pentatonic**: G major pentatonic = E minor pentatonic. Then the one rule that makes improvising sound deliberate: **target notes**. On beat 1 of each bar, land on a note of the current chord. Between those, anything from the scale.

G major pentatonic: G A B D E



F major pentatonic: F G A C D



Example: 4 bars over the Loneliest Girl verse. (Marked notes are targets, chord notes on beat 1.)



Today's rule (pick one, keep it the whole session)

R1 Three notes. Only steps 1, 2, 3. All the interest has to come from rhythm.

R2 Neighbours. All five notes, but only step to the next one; no jumps.

R3 Targets. Beat 1 of every bar lands on a chord note.

R4 Talk and listen. Play 2 bars, rest 2 bars. The rest is part of the music.

R5 One rhythm. Keep the same rhythm every bar, change only the notes.

R6 Full scale. Add steps 4 and 7 as passing notes between pentatonic notes.

The 5 minutes

0:00–1:00

Scale warm-up. Pentatonic up and down, 2 octaves, in today's key.

1:00–2:00

Start the loop. Left hand loops roots or B2 (Drill 5): Em – C – G – Bm in G, or Dm – Bb – F – C in F.

2:00–5:00

Improvise with today's rule. Don't stop for mistakes. Record the whole thing. If 2 bars sound good, write them down: that's a **motif** for Drill 10.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	R1 and R2 over the G loop, left hand on whole notes	70	_____
☐ L2	R3 over the G loop	70	_____
☐ L3	R1–R3 over the F (Scientist) loop	70	_____
☐ L4	Gate. R4 and R5, both keys, left hand playing B2 8ths	80	_____
☐ L5	E minor and D minor pentatonic over the same loops	80	_____
☐ L6	R6 with swing over the lo-fi loop (Drill 8)	75	_____
☐ L7	New key each week: work out its pentatonic with Drill 2	any	_____

Make it harder

Hum first, then find it on the keys. Keep the melody inside 5 notes, then stretch it over 2 octaves. Improvise over the S3 groove from Drill 7.

Song link

Both songs' melodies mostly use these pentatonic notes and land on chord notes at the chord changes. Learning them later will feel familiar.

Build a Tune

One small idea, repeated and changed, becomes an 8-bar melody.

Needs: Drill 9 (L4) **Skills:** motif, variation, question and answer **Gate:** none; keep it in rotation

Why this matters

Melodies are rarely invented note by note. They grow from a **motif**: 3 to 5 notes with a recognisable **shape** (up, up, down...) and **rhythm**. The motif gets **repeated** (so it's memorable) and **varied** (so it isn't boring).

Four variation tools: repeat it exactly; move it up or down to fit the next chord (same shape); keep the rhythm but change the notes; keep the notes but change the rhythm.

Question and answer: a 4-bar phrase that ends away from home (on step 2 or 5) sounds like a question. Repeat it, but end on step 1 or 3, and it sounds like the answer. Give the whole melody one **peak**, its highest note, once. See *Making Music* (DeSantis), "Creating Melodies 1: Contour" p. 178 and "2: Using Motives" p. 182.

Example in C over C – Am – F – G | C – Am – F – C

The 5 minutes

0:00–1:00

Make a motif. 3–5 notes from the pentatonic, one rhythm. Hum it until you can remember it without the keys.

1:00–3:00

Build 4 bars over the loop. Bar 1 motif. Bar 2 same shape, moved to fit the chord. Bar 3 same rhythm, new notes, reach the peak. Bar 4 ends on step 2 or 5: the question.

3:00–4:00

Answer it. Play bars 1–4 again and rewrite the last bar to land on step 1.

4:00–5:00

Record the 8 bars and give the tune a name. Write the notes into a "motif notebook". After a month you'll have 20 ideas to use.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	4-bar question over C – Am – F – G, right hand only	70	_____
☐ L2	Full 8 bars (question + answer)	70	_____
☐ L3	Left hand plays roots or B2 while you play the melody	70	_____
☐ L4	Over the lo-fi loop (Drill 8), swung, arpeggio fill in the rests	70	_____
☐ L5	In minor: A minor or E minor pentatonic, ending on step 1 of the minor key	70	_____
☐ L6	Write a bass line that moves against the melody (melody up, bass down); DeSantis p. 186	70	_____
☐ L7	DAW sketch: melody on a lead synth, chords on a pad, bass on a bass synth	—	_____

Make it harder

Start the motif on an "&" instead of beat 1 (Drill 1). Use two motifs, A and B, in the form A B A B'. Take 2 bars from a song you like, change one thing, and see where it leads.

Song link

Look at the melodies of both songs through this page: find the motif, the repeats, the peak, where the question ends and where the answer ends. You'll learn them faster, and they'll feed your own writing.

The Big Five Progressions

Five chord loops behind a huge share of pop, EDM, lo-fi and anime. Learn them as numbers.

Needs: Drill 4 (L4) **Skills:** progressions, chord function, transposing **Gate:** L3

Why this matters

A **progression** is an order of chords, usually looped. Most songs reuse a few favourites, so learning them pays off twice: you can write with them, and you recognise them in songs you want to learn. Each chord has a job, called its **function**: **I** = home, settled. **vi** = home, but sad. **IV** = lift, moving away. **V** = tension that pulls back to I. **ii** = soft set-up for V. **iii** = bittersweet, in between. A progression is a little story of leaving home and coming back. Learn each loop as **numbers**, then play it in any key by looking up the numbers (Drill 3 table).

	Name	Numbers	C major	G major	F major	Sound, where
P1	Pop axis	I – V – vi – IV	C G Am F	G D Em C	F C Dm B $\flat$	bright pop
P2	Sad axis	vi – IV – I – V	Am F C G	Em C G D	Dm B $\flat$ F C	emotional, EDM
P3	Fifties	I – vi – IV – V	C Am F G	G Em C D	F Dm B $\flat$ C	ballads
P4	Two-five-one	ii7 – V7 – Imaj7	Dm7 G7 Cmaj7	Am7 D7 Gmaj7	Gm7 C7 Fmaj7	jazz, lo-fi
P5	Royal Road	IVmaj7 – V7 – iii7 – vi	Fmaj7 G7 Em7 Am	Cmaj7 D7 Bm7 Em	B $\flat$ maj7 C7 Am7 Dm	J-pop, anime

P1, P2 and P3 use the same four chords in different orders. P2 is also the minor-key favourite of trance and EDM: in A minor it reads i – VI – III – VII.

P1 and P5 in C, voiced the Drill 4 way (right hand moves by steps)

P1 I-V-vi-IV

P5 IVmaj7-V7-iii7-vi7

The 5 minutes

0:00–1:00

P1 in C with the voicing above, saying the numbers out loud: “one, five, six, four”.

1:00–3:00

Progression of the day. Rotate P1 to P5, one per day. Play it in C, then in G or F using the table. Find smooth voicings yourself (top note moves by a step or stays).

3:00–4:00

Give it a groove. Same loop with a Drill 1 rhythm or the Drill 7 S3 groove.

4:00–5:00

Hum over it. Loop the progression and hum a melody. Every progression suggests a different mood; notice which one you like best.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1	P1 and P2 in C, smooth voicings, left-hand root	70	_____
☐ L2	All five in C	70	_____
☐ L3	Gate. All five in G and F from the table	70	_____
☐ L4	Each with a groove (Drill 1 pattern C or Drill 7 S3)	90	_____
☐ L5	P2 in minor keys: A minor, E minor, D minor (i – VI – III – VII)	80	_____
☐ L6	Transpose any loop to D and B $\flat$ without the table, from the numbers only	70	_____
☐ L7	Ear test: name the progression in 5 songs you like (start with the two target songs)	—	_____

Make it harder

Two chords per bar. Play P4 in every key you know. Swap one chord for its colour version (Drill 12): C → Cadd9, G → Gsus4 then G. Write a 4-bar melody over P5 (Drill 10).

Song link

The Scientist verse is P2 without the V (vi–IV–I). **The Loneliest Girl** chorus is IV–I–vi–V, the P1 chords in a new order; its pre-chorus ends IV–I–ii–V, a P4 cousin.

Chord Vocabulary

The chord shapes both songs need, and the nine chord types behind almost every chord symbol.

Needs: Drill 3 (L4) **Skills:** fast chord shapes, chord types, reading chord symbols **Gate:** L3

Why this matters

You don't need hundreds of chords. Every chord symbol is a **root** plus a **recipe** of steps from that root's major scale (Drill 2). C = steps 1-3-5 of C major. Cm = 1-b3-5 (step 3 lowered by a half step). Cmaj7 = 1-3-5-7. With **nine recipes** and any root you can build almost any chord you'll meet in pop, lo-fi or EDM. The nine shapes below are **every chord both target songs use**. They should come instantly, without thinking about the notes. That is what makes songs feel easy.

The nine song chords (R = root, 3 = third, 5 = fifth; right hand fingers 1-3-5)

C

I in C, V in F

D

V in G

F

I in F, IV in C

G

I in G, V in C

Bb

IV in F

Dm

vi in F

Em

vi in G

Am

ii in G, vi in C

Bm

iii in G

Major: 4 + 3 half steps.
 Minor: 3 + 4.
 Black keys: F# in D and Bm, Bb in Bb.

The nine recipes, all on C (the numbers under the notes are the recipe)

C bright	Cm sad	C^{sus2} open	C^{sus4} hanging	Cadd9 sparkly	C^{maj7} dreamy	C⁷ tense	Cm⁷ mellow	C^o uneasy
1 3 5	1 b3 5	1 2 5	1 4 5	1 3 5 9	1 3 5 7	1 3 5 b7	1 b3 5 b7	1 b3 b5

Where you meet them: major and minor everywhere; sus2 and add9 in The Scientist and pop piano; sus4 resolving to major at the end of phrases; maj7 and m7 in lo-fi (Drill 8); 7 on the V chord; dim as a passing chord between two others.

The 5 minutes

- 0:00–1:00** **Shape run.** The nine song chords in random order, eyes on the page, not on your hands. One per beat at 60.
- 1:00–3:00** **Recipe ladder** on the root of the day (rotate C, G, F, D, A, E). Play all nine types and say each sound word.
- 3:00–4:00** **Chord quiz.** Pick a root and a type at random and play it within 2 seconds. Say the notes aloud.
- 4:00–5:00** **Play.** Take P1 from Drill 11 and colour it: Cadd9 – Gsus4 G – Am7 – Fmaj7. Loop it slowly with pedal.

Level ladder — stay on a level for days or weeks; tick it when you play it clean 3 times in a row

☐ L1 The nine song chords, root position, right hand then left hand	60	_____
☐ L2 The nine song chords in all three positions (Drill 4)	60	_____
☐ L3 Gate. Recipe ladder on C, G and F	—	_____
☐ L4 Recipe ladder on D, A, E and Bb	—	_____
☐ L5 Chord quiz on any white-key root in under 2 seconds	—	_____
☐ L6 Slash chords: C/E, G/B, Am/G (chord over a different bass note). Play C – G/B – Am – Am/G – F	70	_____
☐ L7 Black-key roots: Eb, Ab, Db, F#; all 12 major and minor chords	—	_____

Make it harder

Quiz with your eyes closed. Play every recipe in 1st inversion. Read chord symbols from any song chart and build each one from its recipe.

Song link

With these nine shapes automatic, both songs are only a matter of rhythm and melody. For your own tracks, the recipe ladder is a menu of moods: swap a plain chord for sus2, add9 or maj7 and listen.

Song Roadmap

How the drills add up to the two songs. Chords only; learn the melodies from licensed sheet music or a tutorial.

The Scientist (Coldplay) F major (one flat) about 73 BPM 4/4 slow, even, pedalled

Intro / verse Dm7 B $\flat$ (add9) F F – Fsus2 = vi – IV – I – I, repeated

Chorus Moves between B $\flat$ and F (with Fmaj7 / F6 colours) and ends on Csus4 – C = IV – I ... V

Texture Left hand: soft, even 8th-note roots (Drill 5, B2). Right hand: held chords voiced so F and C stay put (Drills 4 and 8). Melody sits on top in F major.

Build-up:

1. Verse loop, left-hand whole-note roots, right-hand shapes from Drill 4. Pedal on each chord. (Drill 4 L4)
2. Left hand switches to B2 8ths. Start at 60 and climb to 73. (Drill 5 L4)
3. Colours: Dm7, B $\flat$ add9, Fsus2 (Drill 8). Chorus: B $\flat$ – F, ending Csus4 → C.
4. Melody: learn the right hand alone first, in phrases, over a backing of your own left hand. Then add the left hand with whole-note roots, then 8ths.
5. Play along with the recording. Sing it while you play, if you like; it helps the phrasing.

The Loneliest Girl (Carole & Tuesday) G major (one sharp) about 112 BPM 4/4 pop groove

Intro / verse Em C G Bm = vi – IV – I – iii

Pre-chorus C G Em D C G Am D = IV – I – vi – V, IV – I – ii – V

Chorus C G Em D = IV – I – vi – V

Texture Left hand on the beat (roots, octaves). Right hand chords on 3-3-2 hits and pushes (Drills 1 and 7). Melody in G major.

Build-up:

1. All three loops with Drill 4 shapes, one chord per bar at 70. Am in the pre-chorus is A-C-E, or C-E-A to stay close to C and G.
2. Drill 7 S3, then S4, on every section. Climb the metronome ladder to 112.
3. Play the form: intro, verse, pre-chorus, chorus. Chord changes without looking down.
4. Melody: right hand alone, slower (about 80), phrase by phrase. Then with left-hand roots only, then with the groove.

Chord charts published online differ in details (sus and add9 chords, chorus turnarounds). Treat these as the skeleton and check against the recording by ear. That is good ear training too.

Progress log (write the date when you tick a level)

Drill	L1	L2	L3	L4	L5	L6	L7
1 Count the Groove							
2 The Major Scale							
3 Chord Staircase							
4 Lazy Hands							
5 The Bass Engine							
6 Broken Chords							
7 Split the Beat							
8 Lo-fi Colours							
9 No Wrong Notes							
10 Build a Tune							
11 Big Five Progressions							
12 Chord Vocabulary							